



Live CDs My Wonder Wall

When I heard the term Live CD, I thought it was a rock band—but very soon, I learnt what a wonderful thing it was. This article not only introduces readers to the concept, but also shows you how to create a Live CD of your own.

A Live CD is a convenient and easy approach for users to try out an operating system and run pre-configured software, or do virtually everything that you can with an installed system. It's a bootable CD-ROM disk that loads an operating system without the need to permanently install it on a storage device like a hard drive.

The origin of the Live CD was not a CD at all, but a bootable floppy disk. Usually, hardware manufacturers and anti-virus developers produced bootable floppy disks for various operating systems, to perform certain administrative tasks that were not possible with the OS already running. For example, anti-virus software needed users to boot their system in a known safe condition, so that any virus infections on the machine would not interfere with virus-testing activity. Hardware manufacturers distributed bootable floppy disks to allow the system user to test the hardware products without OS intervention, and to be sure that the firmware was working correctly.

Now, from the general user's point of view, Live CDs can be used to repair non-working systems. For example, I used it to repair my GRUB boot-loader after a Windows installation. You can also use it to browse the Internet if, for instance, you go to a cyber café, where you're sceptical about whether the computer has key-loggers or other spyware on it, that could steal your personal information. You can just insert a Live CD, and within a short time, you're ready to use the computer with your own, clean and secure operating system! You can even use your pen drive as a storage medium, or can play online/offline games. The best part is that your CD is read-

only, and cannot be tampered with, or infected, so it is far safer than an installed OS.

Now, let's talk about some history. The first Live CD was FM town OS, first released in 1989. The first Linux-based Live CD was Yggdrasil Linux, released in 1992-93, with the aim of describing Linux as plug-and-play, but very soon, its production stopped. Then, in 2000, the Finnix Live CD was born; it is still in production. The most remarkable Linux Live CD is Klaus Knopper's creation, Knoppix—a Debian-derivative Linux distribution. Presently, almost all Linux distributions provide live environments—for example, Fedora, Mandriva, Ubuntu, etc.

How does a Live CD boot?

How does a computer system run from a CD, without the need to permanently install to hard disk? When you press the power button, the first thing that happens is that the BIOS (Basic Input/Output System) performs a POST (Power-On Self-Test). The POST is a built-in diagnostic program that checks computer hardware to ensure that all detected hardware is functioning properly. If everything is all right, it looks into an ordered list of boot devices, and loads an operating system from the first device that offers a bootable image. To make a computer boot from a Live CD, it is often necessary to go into the BIOS set-up screens and change the boot order to put the CD-ROM (or DVD) drive ahead of the hard disk drive (HDD).

Once loaded, the Live CD operating system detects hardware attached to the motherboard, such as PCI

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